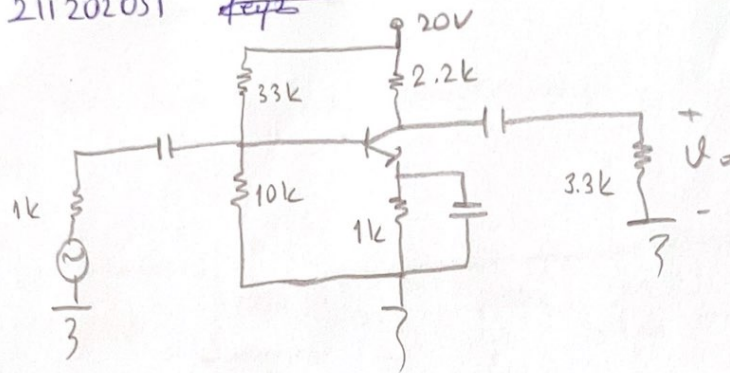
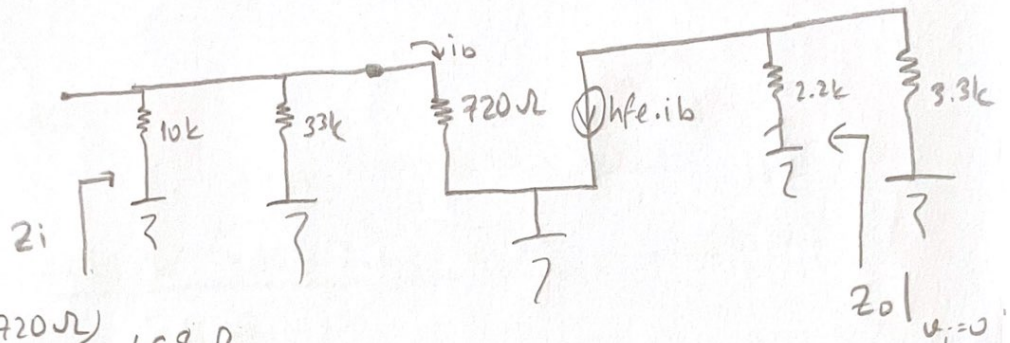


Feyza Baltürk 211202051 ~~4425~~

1) $\beta = 110$
 $h_{ie} = 720 \Omega$



a) $z_i, z_o = ?$



$$z_i = \frac{10k // 33k // 720 \Omega}{7.67k} = 658 \Omega$$

$$z_o = 2.2k$$

$$z_i = 658 \Omega$$

b) $A_v, A_{vs}, A_i = ?$

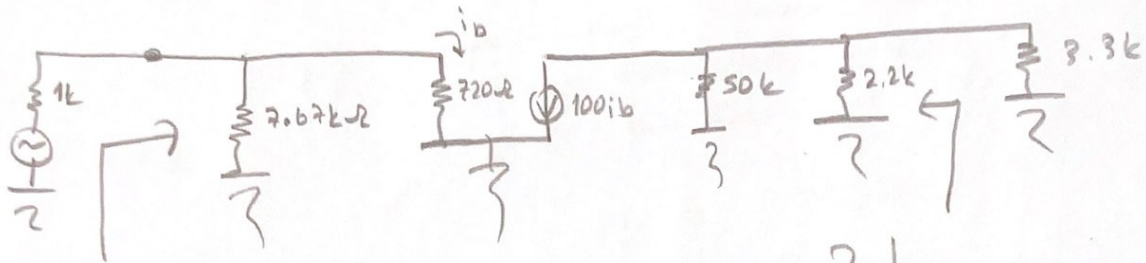
$$A_v = \frac{U_o}{U_i} = \frac{-h_{fe} \cdot i_b \cdot (2.2k // 3.3k)}{720 \Omega \cdot i_b} = \frac{-100 \cdot 1.32k \Omega}{720 \Omega} = -183.33$$

$$A_{vs} = A_v \cdot \frac{z_i}{z_i + R_s} = -183.33 \cdot \frac{658 \Omega}{1658 \Omega} = -72.75$$

$$A_i = \frac{i_o}{i_i} = -A_v \cdot \frac{z_i}{R_y} = +183.33 \cdot \frac{658 \Omega}{3300 \Omega} = 36.55$$

c) $h_{oe} = 20 \mu S$ z_i z_o ve A_v değerlerini hesaplayınız.

$$r_o = \frac{1}{20 \mu S} = 50 k$$



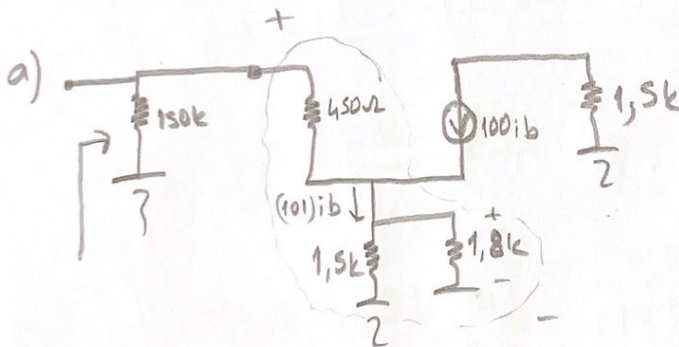
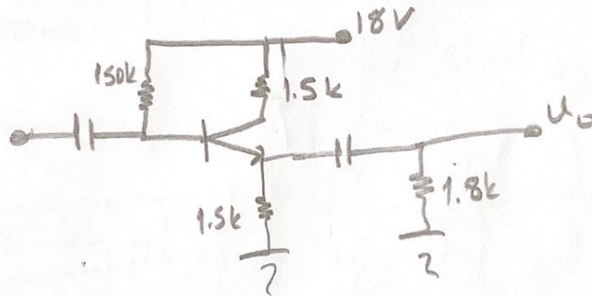
z_i değişmez

$z_o |_{u_s=0} \text{ için}$

$$1.28k \parallel (50 \parallel 32)k = 2.10k$$

$$A_v = \frac{u_o}{u_i} = \frac{-100 i_b (3.3k \parallel 2.2k \parallel 50k)}{720 \Omega \cdot i_b} = -177.78$$

2) Yandaki devrede
 $\beta = 100$ $h_{ie} = 450 \Omega$



a) $z_i, z_o = ?$ $(1.5 \parallel 1.8)k = 0.8k$

$$z_i = R_B \parallel z_B$$

$$z_B = \frac{u_i}{i_b} = \frac{450 i_b + 0.8k \cdot 101 \cdot i_b}{i_b}$$

$$z_B = 81.2k$$

$$z_i = (150 \parallel 81.2)k = 52.68k$$

b) $A_v, A_i = ?$

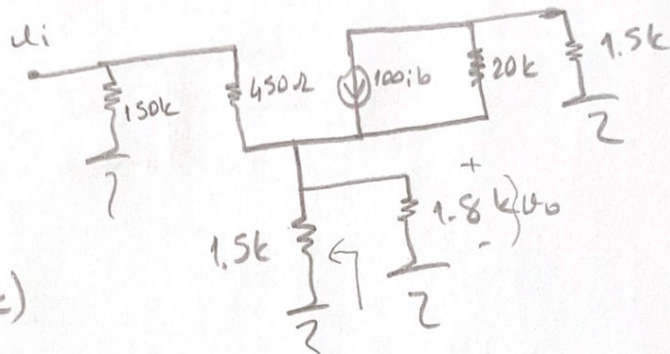
$$A_v = \frac{u_i}{u_o} = \frac{z_B \cdot i_b}{0.8k \cdot 101 \cdot i_b} = \frac{81.2k}{0.8k \cdot 101} = 1 //$$

$$A_i = -A_v \cdot \frac{z_i}{R_y} = -1 \cdot \frac{52.68k}{1.8k} = -29.3$$

Feyza Baltürk 211202051

c) $h_{oe} = 50 \mu S$ ise;

$Z_i, Z_o, A_v = ?$



$$Z_i = 150k \parallel Z_B$$

$$Z_B = 450 \Omega \cdot i_b + 0,8k \cdot (i_b + I_C)$$

I_C 'yi bulmak için

1. çevre;

$$21,5k \cdot I_C + (0,8)(i_b + I_C) = 2000k i_b$$

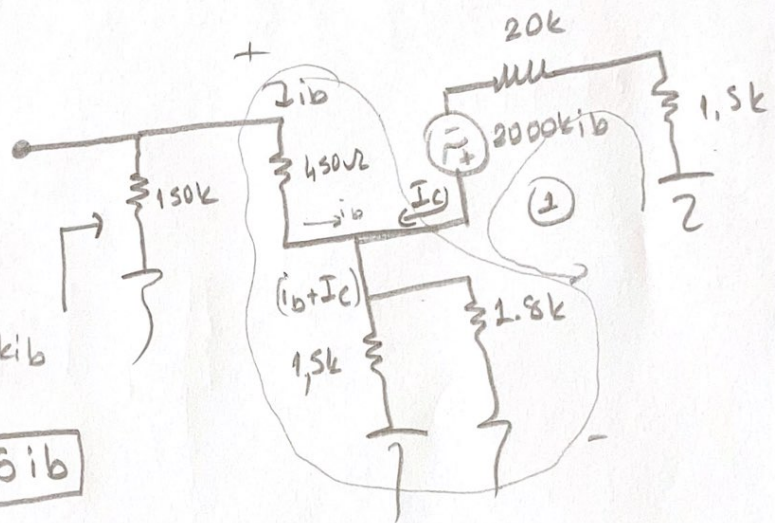
$$(21,5 + 0,8)k \cdot I_C + 0,8k i_b = 2000k i_b$$

$$I_C = \frac{(2000 - 0,8)k}{(22,3)k} = 89,65 i_b$$

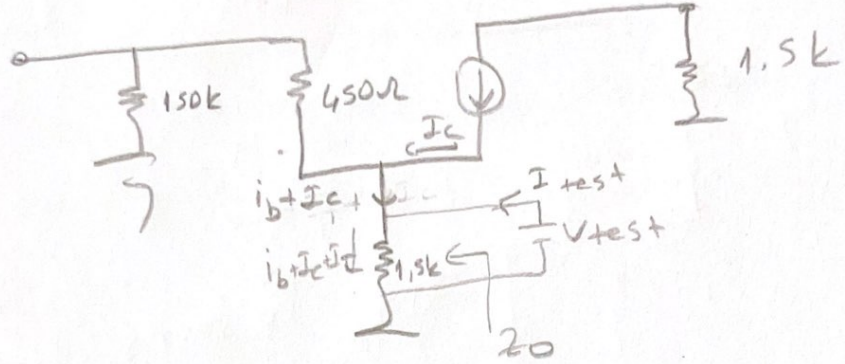
$$Z_B = \frac{u_i}{i_b} = \frac{450 \Omega \cdot i_b + (0,8k) \cdot 90,65 i_b}{i_b} = 72,9k \Omega = Z_B$$

$$Z_i = 150k \parallel 72,9k = 49k \Omega = Z_i$$

Z_o için;



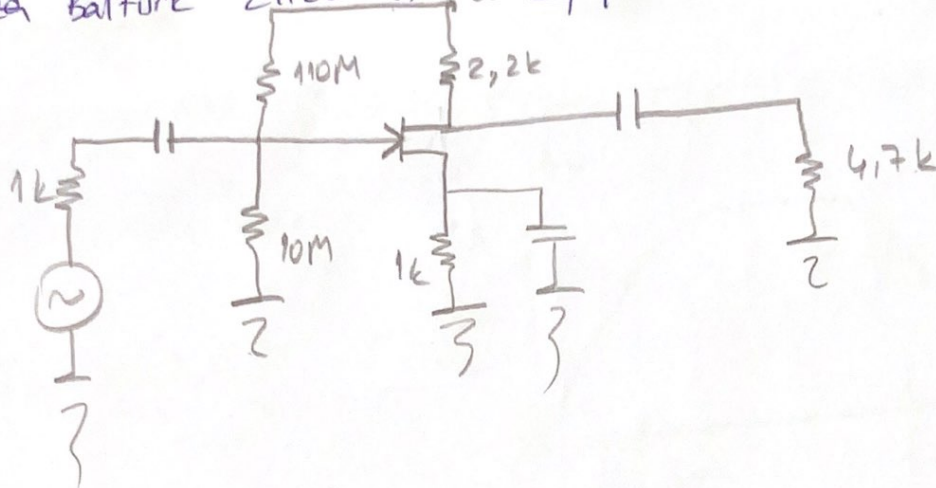
fevza Baltürk 211202051 ~~fevza~~



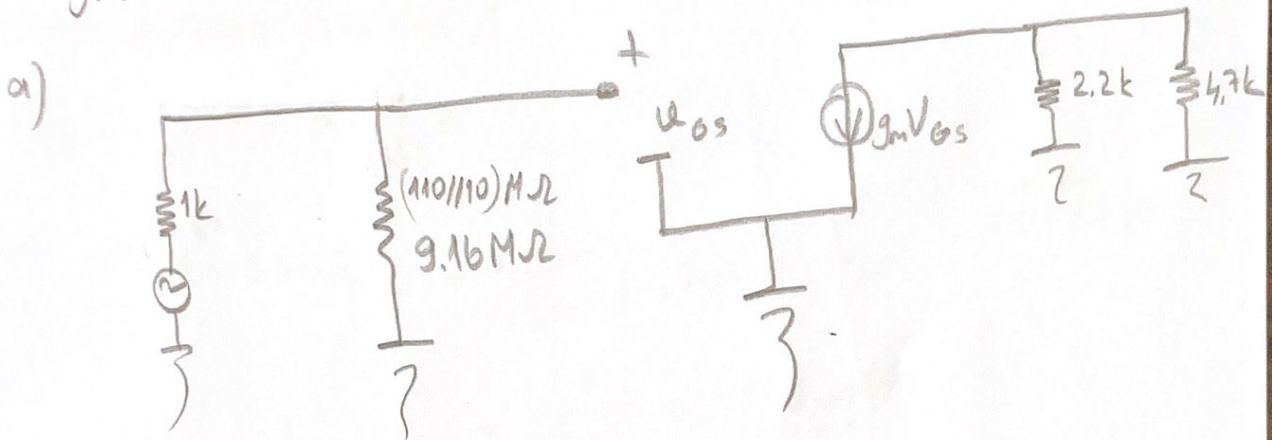
faiz - " " "

Feyza Baltürk 211202051
 Feyza Baltürk 211202051 20v ~~Feyza~~

3)



$g_m = 3,56 \text{ mS}$ i_{in}



b) $z_i = R_G = 9,16 \text{ M}\Omega$

c) $z_o = R_D = 2,2 \text{ k}\Omega$

d) $A_v = -g_m (R_D) = -3,56 \text{ mS} \cdot 2,2 \text{ k}\Omega = -7,832$

e) $A_i = \frac{i_o}{i_i} = -A_v \cdot \frac{z_i}{R_L} = +7,832 \cdot \frac{9,16 \text{ M}\Omega^{10^3}}{4,7 \text{ k}\Omega} = \underline{\underline{15264}}$

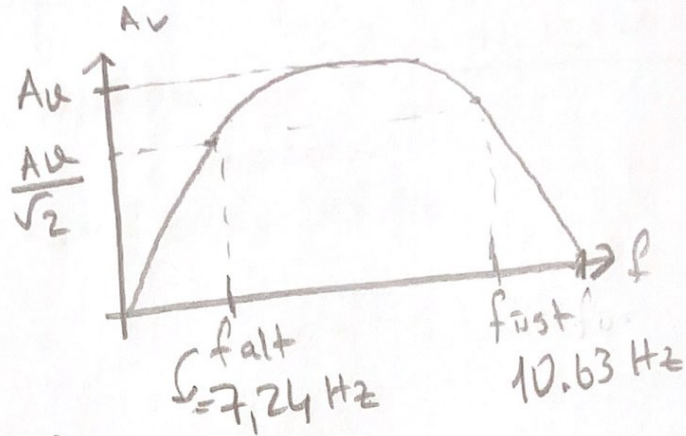
Feyza Baltürk 211202051 ~~feyza~~

4) yükselteç için band genişliğini

hesaplayarak kazanç frekans eğrisini çiziniz.

$$BW = f_{üst} - f_{alt}$$

$$A_v = - \frac{h_{fe} \cdot R}{h_{ie}}$$



alçak frekans analizi ;

$$C_g \text{ 'nin etkisi} = f_{LG} = \frac{1}{2\pi(R_k + Z_i) \cdot C}$$

$$R_k = 1k \\ Z_i = 9.16M\Omega$$

$$= \frac{1}{2\pi(1k + 9.16M\Omega) \cdot 10\mu F} = \frac{1}{183.22\pi}$$

$$= \frac{1}{575.6} = 1.73 \text{ mHz}$$

C_D 'nin etkisi

$$R_y = 4.7k$$

$$R_D = 2.2k$$

$$f_{LD} = \frac{1}{2\pi(R_D + R_y) \cdot C_D} = \frac{1}{2\pi \cdot (6.9k) \cdot 10\mu F}$$

$$= \frac{1}{0.434} = 2.30 \text{ Hz}$$

C_E 'nin etkisi

$$f_{LS} = \frac{1}{2\pi \cdot C_E \cdot R_{E_S}}$$

$$R_{E_S} = R_S \parallel \frac{1}{g_m}$$

$$= 1k \parallel \frac{280.9}{3.56m}$$

$$= \frac{1}{0.138} = 7.24 \text{ Hz}$$

$$= 220\Omega$$

$$f_{alt} = \max(f_D, f_G, f_S)$$

$$f_{alt} = 7.24 \text{ Hz}$$

Feyza Baltürk 211202 051 ~~fezo~~

f_{dst} için ;

yüksek frekans

$$R_i = R_k \parallel R_G \approx R_k = 1k$$

$$f_i = \frac{1}{2\pi C_i R_i} = \frac{1}{2\pi 10MF \cdot 1k} = \frac{1}{0,06} = \underline{16,12 \cdot 29}$$

$$f_o = \frac{1}{2\pi C_o R_o}$$

$\downarrow$
10MF

$$R_o = R_D \parallel R_y \rightarrow 2,2k \rightarrow 4,7k$$

$$(2,2 \parallel 4,7)k = \underline{1,5k}$$

$$\frac{1}{2\pi \cdot 10MF, 1,5k} = \frac{1}{0,094} = \underline{10,63}$$

$$f_{dst} = \min(f_i, f_o) = 10,63$$